The Kuramoto Engine

Physics and Mathematics of the Cardiac Synchronization Index

Technical reference for the NEUROCARDIA application

Version 0.2 — Revision

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Revision history. This is revision 0.2 of *The Kuramoto Engine*. Revision 0.1 (October 14, 2026) specified the model, the numerical scheme, and the parameter choices, and stated $K = 2$ as an empirical choice. Revision 0.2 adds the result of the coupling-strength sweep: $K = 2$ is not merely the value that produces a diagnostic residual, it is the only value in the tested range where the residual carries information at all. The revision also notes that the trajectory of the order parameter is not robust to the frequency distribution at any value of K where the metric has dynamic range. The mathematical treatment of the model is unchanged.

Abstract

This document specifies the mathematical model, the numerical methods, and the parameter choices underlying the Cardiac Synchronization Index (CSI) computed by the NEUROCARDIA application. The CSI is the order parameter of a Kuramoto coupled-oscillator network driven by the RR interval series of a single-lead ECG recording. We present the model, the discretization scheme, the treatment of the natural frequency distribution, the phase-reset mechanism, and the derivation of the four early warning signal metrics that accompany the CSI. Section 4.3 reports the coupling-strength sweep and the narrow window in which the metric is diagnostic. Section 4.2 discusses the sensitivity of the trajectory to the frequency distribution. The document is self-contained: it assumes no prior familiarity with the Kuramoto model.

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1 Introduction

The Kuramoto model is the canonical mathematical description of synchronization in a population of coupled phase oscillators. It was introduced by Yoshiki Kuramoto in 1975 [1] as a minimal model of collective behavior, and it has since been applied across physics, biology, and neuroscience [2, 3]. The appeal of the model is that it captures the essential phenomenon — oscillators settling into or falling out of phase lock — without requiring detailed specification of the underlying dynamics.

This document describes one specific application of the Kuramoto model: the analysis of cardiac rhythm from the RR interval series of a single-lead ECG. The sinoatrial node is a population of coupled pacemaker cells whose collective firing produces the cardiac rhythm. Treating the RR tachogram as a phase signal and applying the Kuramoto model to it yields a time-varying order parameter that we call the Cardiac Synchronization Index (CSI). The CSI is a measure of how regularly the heart is beating, expressed in the phase domain rather than the time or frequency domain.

The document is organized as follows. Section 2 presents the Kuramoto model and the order parameter. Section 3 describes the RR-to-phase mapping that connects the model to the ECG signal. Section 4 specifies the network parameters, including the frequency distribution and the coupling strength. Section 5 presents the numerical integration scheme. Section 6 derives the early warning signal metrics. Section 7 discusses mathematical properties of the model. Section 8 identifies the model's limitations.

2 The Kuramoto Model

2.1 The phase oscillator

Consider a system of N coupled oscillators. Each oscillator i is described by a single scalar, its phase $\theta_i \in \mathbb{R}/2\pi\mathbb{Z}$. The phase advances monotonically in time; the rate of advance is the oscillator's instantaneous frequency. When the oscillator is isolated, the rate is a constant called the natural frequency ω_i.

The choice to describe an oscillator by its phase rather than by a full state vector is a deliberate simplification. It is justified when the oscillator's amplitude is either fixed or irrelevant to the question being asked, and when the only observable of interest is timing. For coupled pacemaker cells driving a cardiac rhythm, both conditions hold: the observable that matters is when the next beat occurs, not the detailed ionic dynamics that produced it.

2.2 The coupling

When two oscillators are coupled, each one's instantaneous frequency is perturbed by the phase of the other. A natural form for this perturbation is the sine of the phase difference. This choice has three properties that make it the canonical one:

1. *Antisymmetry.* If oscillator j advances the phase of oscillator i by $\sin(\theta_j - \theta_i)$, then i advances j by $\sin(\theta_i - \theta_j) = -\sin(\theta_j - \theta_i)$, the exact opposite. The two oscillators influence each other equally and oppositely. No energy is created or destroyed by the coupling.

2. *Smoothness.* The sine is infinitely differentiable and bounded. It cannot produce abrupt jumps in frequency.

3. *2π-periodicity.* The coupling depends only on the phase difference modulo 2π. Two oscillators at phase difference 0.1 and at phase difference $0.1 + 2\pi$ are coupled identically.

More general coupling functions are possible and are studied in the literature. The sine coupling is the simplest that satisfies the requirements above, and it is the form used here.

2.3 The governing equation

The Kuramoto model is the system of N coupled differential equations

$$\frac{d\theta_i}{dt} = \omega_i + \frac{K}{N}\sum_{j=1}^{N}\sin(\theta_j - \theta_i), \quad i = 1, \ldots, N, \tag{1}$$

where $K \geq 0$ is the coupling strength. The factor $1/N$ normalizes the sum so that the total influence on oscillator i does not grow without bound as N increases. With this normalization, the coupling term is the average of the sine of the phase differences between i and every other oscillator.

The equation has a clear physical reading. The phase of oscillator i advances at its natural frequency ω_i, plus a correction from the coupling. If i is lagging behind the population on average, the sum of sines is

positive and i speeds up. If i is leading, the sum is negative and i slows down. The coupling pulls oscillators toward each other.

2.4 The order parameter

The degree to which the population is synchronized is quantified by the order parameter

$$r = \left| \frac{1}{N} \sum_{j=1}^{N} e^{i\theta_j} \right|, \tag{2}$$

where $e^{i\theta_j}$ is the point on the unit circle corresponding to phase θ_j. The order parameter has a direct geometric interpretation: it is the magnitude of the mean of N unit vectors, each pointing at the phase of one oscillator.

If all oscillators have the same phase, the mean of their unit vectors is a unit vector, and $r = 1$. If the phases are uniformly distributed around the circle, the unit vectors sum to zero, and $r = 0$. For a finite population of N oscillators with uniformly random phases, the expected value of r is $1/\sqrt{N}$.

The order parameter is bounded in $[0, 1]$ and is invariant to a global phase shift: adding a constant to every θ_j does not change r.

2.5 The critical coupling

For a given distribution of natural frequencies $g(\omega)$, the Kuramoto model exhibits a phase transition at a critical coupling strength K_c. Below K_c, the oscillators drift incoherently and $r \to 0$ in the thermodynamic limit. Above K_c, a subset of oscillators locks to a common frequency and r rises toward a positive value.

For a symmetric, unimodal distribution $g(\omega)$, the critical coupling has the closed form

$$K_c = \frac{2}{\pi g(0)}, \tag{3}$$

where $g(0)$ is the density of the frequency distribution at its center. Distributions with a sharper peak at zero have smaller K_c; distributions that are flatter at zero have larger K_c.

This closed form does not apply to the frequency distribution used here (see Section 4), but the qualitative behavior holds: there is a threshold below which the network is incoherent and above which it synchronizes. The precise value of the threshold depends on the distribution.

3 RR-to-Phase Mapping

The Kuramoto model describes the evolution of phases. The ECG does not measure phases directly. We construct a phase signal from the RR interval series as follows.

Let $\{R_k\}_{k=1}^{M}$ be the intervals between successive detected R-peaks, in seconds. Let $\overline{R}$ be their mean. Define the cumulative phase at beat k as

$$\phi_k = 2\pi \sum_{j=1}^{k} \frac{\overline{R}}{R_j} \pmod{2\pi}. \tag{4}$$

Each beat advances the cumulative phase by $2\pi\overline{R}/R_j$. A beat at the mean rate ($R_j = \overline{R}$) advances the phase by exactly 2π, returning to the same point on the circle. A shorter interval advances the phase by more than 2π; a longer interval advances it by less.

The choice of $\overline{R}$ as the reference is deliberate. It makes the phase trajectory scale-invariant: a subject whose mean RR is 0.6 seconds (100 BPM) and a subject whose mean RR is 1.2 seconds (50 BPM) produce comparable phase trajectories, both advancing by 2π per mean beat. The CSI measures synchronization, not rate. Rate is reported separately as heart rate in beats per minute.

The cumulative phase is defined modulo 2π. The natural choice is to wrap at the boundary of the unit circle, mapping every phase into $[0, 2\pi)$.

4 Network Construction

4.1 Number of oscillators

The network has $N = 16$ oscillators. This is a modeling choice. The choice is not derived from cardiac physiology; it matches the dimension used in the reference implementation from which the parameterization was inherited. A principled choice — for example, derived from the number of identifiable frequency components in the RR spectrum, or from the number of pacemaker cell clusters in the sinoatrial node — would be preferable, and the sensitivity of the CSI to this choice is a subject for future work.

The incoherent baseline for $N = 16$ is $1/\sqrt{16} = 0.25$. Any CSI value meaningfully above 0.25 indicates partial synchronization.

4.2 Natural frequency distribution

The natural frequencies ω_i are drawn from the non-trivial zeros γ_i of the Riemann zeta function. The first 24 such zeros are tabulated in Table 1. For a network of $N = 16$ oscillators, the first 16 zeros are used.

14.134725	21.022040	25.010858	30.424876	32.935062	37.586178
40.918719	48.005151	49.773832	52.970321	56.446248	59.347044
60.831779	65.112544	67.079811	69.546402	72.067158	75.704691
77.144840	79.337375	82.910381	84.735493	86.970000	87.425275

Table 1: The first 24 non-trivial Riemann zeros. Used as the raw frequencies of the network before normalization.

The raw frequencies are then normalized:

$$\omega_i = \frac{\gamma_i - \bar{\gamma}}{\sigma_\gamma}, \tag{5}$$

where $\bar{\gamma}$ and σ_γ are the mean and standard deviation of the first 16 Riemann zeros.

Two points deserve emphasis.

1. The Riemann zeros are not claimed to have cardiac significance. They are a deterministic, fixed sequence with a mathematically determined distribution. Any other deterministic sequence with comparable spread would serve the same role. The choice reflects the reference implementation rather than a physiological argument.

2. The normalization is required for numerical tractability. Section 5 discusses the integration scheme. The unnormalized frequencies span a range of order 10^2, which would require an impractical step size for stable Euler integration. The normalized frequencies span a range of order 1, which the Euler scheme resolves at a tractable step.

Sensitivity to the frequency distribution. The trajectory of the order parameter depends on the choice of frequency distribution. A sensitivity analysis across five distributions (Riemann zeros, uniform random, Gaussian, physiological scale, and linear spacing) shows that the mean Pearson correlation between the CSI trajectory under the Riemann-zero configuration and under any alternative configuration is at most 0.17 at $K = 2$, and reaches the pre-registered threshold of 0.80 only for the linear-spacing distribution at $K = 30$, where the network is saturated. The absolute value of the CSI is therefore not robust to the frequency distribution at any coupling strength where the metric has dynamic range. The residual diagnostic (Section 7) is partially robust, but not to the extent that revision 0.1 of the companion paper claimed. We defer the full analysis to the companion documents and note here only that the choice of frequency distribution is part of the definition of the CSI, not an incidental parameter.

4.3 Coupling strength

The coupling strength is set to $K = 2$. This value is chosen empirically rather than from the critical coupling formula, and it matters: the behavior of the metric depends critically on this choice. At $K = 2$ the network partially synchronizes after each R-peak reset without saturating, so the order parameter trajectory retains dynamic range. At $K \geq 5$ the network locks completely after each reset and the order parameter sits near 1.0 for all input rhythms, destroying sensitivity. At $K \leq 1$ the network is under-coupled and the residual diagnostic information inverts in sign.

The useful range is narrow and centered on $K = 2$. The companion sensitivity analysis and residual papers report the full coupling sweep. The quantitative summary is:

- $K \leq 1$ **(under-coupled).** The residual separates sinus from AF with opposite sign: AUC 0.17 at $K = 0.5$ and 0.21 at $K = 1.0$ (sign-inverted accuracies of 83% and 79%). The network is too weakly coupled to reflect the phase perturbations from the RR series.
- $K = 2$ **(intermediate).** The residual separates sinus from AF with AUC 0.89, $\rho = +0.68$, $p = 0.002$ on the pilot corpus. This is the useful regime.
- $K \geq 3$ **(saturated).** The residual AUC drops to 0.59–0.72 and the metric becomes indistinguishable from noise. The network locks completely after each reset and stays locked until the next beat, so the order parameter does not respond to the RR series.

The choice of K is part of the metric's definition, not a physical postulate about the sinoatrial node.

4.4 Coupling topology

The default coupling is all-to-all with weight $1/N$. This is the classical Kuramoto configuration. Every oscillator couples to every other with equal weight, and the topology reflects the assumption that the sinoatrial pacemaker cells are densely interconnected.

Three alternative topologies are available in the implementation:

- **Nearest-neighbor.** Each oscillator couples to its two immediate neighbors. Produces traveling waves.
- **Small-world.** A ring with a small number of random long-range links. Balances local clustering with global connectivity.
- **Star.** One hub oscillator couples to all leaves. Concentrates synchronization through a single node.

The topology is a modeling choice. The default is all-to-all.

5 Numerical Integration

5.1 The Euler scheme

The Kuramoto equation (1) is a system of first-order ordinary differential equations with no closed-form solution for $N > 2$. We integrate numerically using the explicit Euler method:

$$\theta_i(t + \Delta t) = \theta_i(t) + \Delta t \cdot \left[\omega_i + \frac{K}{N} \sum_{j=1}^{N} \sin(\theta_j(t) - \theta_i(t)) \right], \tag{6}$$

where Δt is the integration step. The phases are updated simultaneously: the right-hand side is evaluated at the current time t for all i, then all updates are applied.

The choice of the Euler method is deliberate. Higher-order methods (Runge-Kutta, for instance) are more accurate per step, but the Euler method is stable for the step sizes used here and its behavior is easy to reason about. The accuracy of the trajectory is not the objective; the value of the order parameter is, and that quantity is robust to moderate integration error.

5.2 Step size

The step size must be small enough that the phase advance per step is much less than 2π. Otherwise the sine evaluation aliases: the phase advances past one full rotation during a single step, and the update effectively samples a random point on the circle.

With normalized frequencies of order 1 and coupling $K = 2$, the maximum instantaneous phase velocity is of order 2 rad/s. A step of $\Delta t = 0.05$ s advances the phase by up to 0.5 radians per step, well under the aliasing threshold.

The step size is fixed at 0.05 seconds. For a recording with mean RR interval $\overline{R}$, the number of integration steps per beat is

$$n_{\text{steps}} = \left\lfloor \frac{\overline{R}}{\Delta t} \right\rfloor, \tag{7}$$

clamped to the range $[1, 200]$. For a mean RR of 0.83 seconds (72 BPM), this gives 16 steps per beat.

5.3 Phase reset

At each detected R-peak, the phase of a designated heart oscillator (index 0) is reset to the cumulative phase ϕ_k defined in equation (4). The other $N - 1$ oscillators are not reset; they continue their evolution from the current state.

The reset is the mechanism by which the network is coupled to the cardiac signal. Without it, the network would evolve as a free-running oscillator system with no relationship to the subject's rhythm. With it, each beat injects a phase impulse into the network, and the network's response — how it re-synchronizes, how quickly, and with what characteristic fluctuations — is the diagnostic signal.

5.4 Sampling the order parameter

After each integration step, the order parameter r is evaluated using equation (2) and appended to a trajectory. For a recording with M beats, the trajectory has approximately $M \cdot n_{\text{steps}}$ samples.

The CSI reported for the recording is the mean of the trajectory:

$$\text{CSI} = \bar{r} = \frac{1}{T} \sum_{t=1}^{T} r(t), \tag{8}$$

where T is the trajectory length.

6 Early Warning Signals

The CSI is a scalar summary of the trajectory $r(t)$. The trajectory itself carries more information than its mean. Four statistical moments of $r(t)$ are computed and reported alongside the CSI, as early warning signals of impending rhythm transitions.

6.1 Variance

The sample variance of the trajectory is

$$\mathrm{Var}[r] = \frac{1}{T-1}\sum_{t=1}^{T}(r(t)-\bar{r})^2. \tag{9}$$

A rise in the variance of an order parameter is a classic signature of critical slowing down: as a system approaches a bifurcation, it recovers more slowly from perturbations and therefore exhibits larger fluctuations [4].

6.2 Skewness

The sample skewness is

$$\mathrm{Skew}[r] = \frac{T}{(T-1)(T-2)}\sum_{t=1}^{T}\left(\frac{r(t)-\bar{r}}{\sigma_r}\right)^3, \tag{10}$$

where $\sigma_r = \sqrt{\mathrm{Var}[r]}$. The skewness is a measure of asymmetry in the distribution of $r(t)$. It reflects the extent to which the trajectory spends time at values above its mean versus below.

6.3 Excess kurtosis

The excess kurtosis is

$$\mathrm{Kurt}[r] = \frac{1}{T}\sum_{t=1}^{T}\left(\frac{r(t)-\bar{r}}{\sigma_r}\right)^4 - 3. \tag{11}$$

The excess kurtosis measures the weight of the tails of the distribution relative to a Gaussian. Positive excess kurtosis indicates intermittent bursts of incoherence: the trajectory spends most of its time near its mean but occasionally excursions to extreme values.

6.4 Lag-1 autocorrelation

The lag-1 autocorrelation is

$$\rho_1[r] = \frac{\sum_{t=2}^{T}(r(t)-\bar{r})(r(t-1)-\bar{r})}{\sum_{t=1}^{T}(r(t)-\bar{r})^2}. \tag{12}$$

The lag-1 autocorrelation approaches 1 as the system recovers more slowly from perturbations. Together with the variance, it is one of the most reliable early warning signals identified in the literature [4].

7 Mathematical Properties

7.1 Invariance to a common frequency offset

Adding a constant c to every natural frequency in equation (1) is equivalent to applying a global rotation to all phases. To see this, let $\tilde{\theta}_i(t) = \theta_i(t) - ct$. Then

$$\frac{d\tilde{\theta}_i}{dt} = \frac{d\theta_i}{dt} - c = (\omega_i - c) + \frac{K}{N}\sum_{j=1}^{N}\sin(\tilde{\theta}_j - \tilde{\theta}_i), \tag{13}$$

because $\theta_j - \theta_i = \tilde{\theta}_j - \tilde{\theta}_i$. The transformed phases satisfy the same Kuramoto equation with shifted frequencies. The order parameter, equation (2), is invariant to a global rotation, so r is unchanged.

This property is what justifies the frequency normalization of equation (5). Subtracting the mean of the frequencies does not change the relative dynamics. Scaling by the standard deviation also does not change the relative dynamics, up to a time reparameterization: rescaling all frequencies by s is equivalent to rescaling the time axis by $1/s$.

7.2 The uniform distribution is a fixed point

Let all phases be equally spaced around the circle:

$$\theta_i = \frac{2\pi i}{N}, \quad i = 0, 1, \ldots, N-1. \tag{14}$$

For any i, the sum of sines in equation (1) is

$$\sum_{j=0}^{N-1}\sin(\theta_j - \theta_i) = 0, \tag{15}$$

because the pairs θ_j and $\theta_{j'}$ that are symmetric around θ_i contribute opposite sine values. The right-hand side of equation (1) is therefore just ω_i for every i.

If all ω_i are equal, no oscillator moves and the network remains in the uniform configuration forever, at any coupling strength. In the thermodynamic limit $N \to \infty$, this configuration is unstable above K_c: any perturbation grows exponentially. But for finite N, and starting exactly at the uniform distribution, the network is a fixed point.

This is a mathematical property, not a bug. Its practical consequence is that any test of the coupling mechanism must use an initial condition that is not perfectly symmetric. In the CSI pipeline, this property is irrelevant, because the phase resets at each R-peak are inherently non-symmetric.

7.3 Dependence on the number of oscillators

The order parameter is bounded by $r \leq 1$ for any N. The expected value of r for N oscillators with uniformly random phases is

$$\mathbb{E}[r] = \frac{\sqrt{\pi}}{2\sqrt{N}} \cdot \frac{\Gamma(N)}{\Gamma(N+1/2)} \approx \frac{1}{\sqrt{N}} \quad \text{for large } N. \tag{16}$$

For $N = 16$, the incoherent baseline is approximately 0.25. CSI values meaningfully above this baseline indicate partial synchronization.

7.4 Residual information beyond RR variability

The residual of the CSI is defined as the component orthogonal to the coefficient of variation of the RR series. The companion paper reports the full analysis; here we note the mathematical result that makes the residual non-trivial: the CSI is not a linear function of the RR series coefficient of variation, so the OLS residual is not identically zero. At $K = 2$, the residual carries information about the atrial state that the coefficient of variation does not, though the size of that information depends on the frequency distribution (Section 4) and on the coupling strength (Section **??**).

8 Limitations

8.1 The model is a simplification

The Kuramoto model describes a population of coupled phase oscillators with no amplitude, no spatial structure, and sine coupling. The sinoatrial node is none of these things: it has spatial extent, it has finite amplitude dynamics, and the coupling between pacemaker cells is not necessarily sinusoidal. The Kuramoto model is a minimal description, chosen because it captures the essential phenomenon (phase locking) with the fewest parameters.

Whether the minimal model is sufficient for the diagnostic purpose is an empirical question. If the CSI discriminates clinical states and predicts transitions, the minimality is justified. If it does not, a more detailed model is needed. Revision 0.2 of the companion residual paper narrows the claim: the residual carries information beyond standard HRV, but the information is modest and depends on the frequency distribution.

8.2 The frequency distribution is arbitrary

The choice to use the Riemann zeros as the frequency distribution is inherited from the reference implementation. It is not derived from cardiac physiology. The sensitivity analysis shows that the trajectory of the CSI is not robust to this choice at any useful coupling strength, and that the residual is diagnostic for some distributions but not others. The frequency distribution is therefore part of the definition of the CSI, not an incidental parameter.

8.3 The coupling strength is a free parameter

$K = 2$ is the only value in the tested range where the residual separates sinus from AF. The useful range is narrow: below $K = 1$ the residual inverts in sign, and above $K = 3$ the network saturates. The choice is empirical, not derived. The sensitivity analysis reports the full sweep.

8.4 The single-lead constraint

The ECG analyzed is single-lead. No spatial information is available. The CSI is a temporal metric: it describes the evolution of synchronization over time, not its distribution over the atrium. A multi-lead device would allow a spatial extension, but that is out of scope for this application.

9 Conclusion

The Kuramoto model provides a mathematically rigorous framework for describing phase synchronization in coupled systems. Applied to the RR tachogram of an ECG recording, it yields the Cardiac Synchronization Index, a phase-domain measure of how regularly the heart is beating. The model's parameters are

documented, its numerical integration is described, and its limitations are stated. The coupling strength and the frequency distribution are the two parameters that determine whether the metric is diagnostic; both are reported in the companion documents.

The CSI is a research metric. Its clinical validity depends on the outcomes of the validation protocols described in the companion white paper and the sensitivity analysis. The mathematics in this document establishes that the metric is well-defined and computable; it does not establish that the metric is clinically useful.

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